

AI Full■Stack Project Builder Guide

How to build 90%+ of a project using AI with proper documentation, prompts, and free AI tools

1. Overview

If you want AI to generate a full stack project (frontend, backend, database, APIs, deployment), you cannot just write a simple prompt like 'build a website'.

Professional AI builders give structured project documents to the AI. These documents act like a blueprint for the system.

Once AI reads these documents, it can generate:

- Frontend UI
- Backend APIs
- Database schema
- Authentication
- Deployment configuration

Usually AI can build around 85■92% of the system automatically. The remaining work is debugging, security fixes, and optimization.

2. Required Project Documents

Create a folder called:

/project-docs

Inside it create these files:

PRD.md
Features.md
UIUX.md
TechStack.md
Database.md
API.md
Architecture.md
Security.md
Deployment.md
AI_Instructions.md

Each document gives AI a different type of information about the system.

3. PRD (Product Requirement Document) Format

Purpose:

Explain the product idea and problem.

Example:

Project Name: AI Resume Analyzer

Problem:

Students do not know if their resume passes ATS filters.

Target Users:

Students and job seekers

Core Features:

- Resume upload
- AI analysis
- ATS score
- Improvement suggestions

User Flow:

Landing Page → Signup → Upload Resume → Analysis → Report

4. Features Document Example

Feature List:

Authentication

User signup and login using email and password

Dashboard

User can see uploaded resumes and reports

Resume Upload

User uploads PDF or DOC file

AI Analysis

System analyzes resume and generates score

Reports

User sees improvement suggestions

5. Database Schema Example

Users Table

id
name
email
password
created_at

Resumes Table

id
user_id
file_url
score
suggestions
created_at

6. API Specification Example

POST /api/signup

POST /api/login

POST /api/upload-resume

GET /api/report/:id

GET /api/user-reports

7. System Architecture Example

Frontend → Next.js

Backend → Node.js + Express

Database → PostgreSQL

Storage → AWS S3

AI Service → Python microservice

8. Security Requirements

Always instruct AI to implement:

- Input validation
- JWT authentication
- Rate limiting
- Environment variables
- SQL injection protection
- Secure password hashing

9. AI Prompt to Generate the Project

Prompt:

Read all files inside /project-docs.

Generate a production-ready full stack project based on these documents.

Requirements:

- modular architecture
- secure APIs
- scalable structure

10. Free AI Coding Stack

You can build the entire project using free AI tools.

AI IDE:

Cursor
Windsurf

AI Models:

Claude
OpenAI Codex

Deployment Platforms:

Vercel
Railway
Render

11. How to Use Cursor (Step-by-Step)

1. Install Cursor IDE.
2. Open your project folder.
3. Create a folder called /project-docs.
4. Add all documentation files.
5. Open AI chat inside Cursor.

6. Paste the master prompt.
7. Ask AI to generate project modules step by step.

Example prompts:

Generate backend API structure.

Generate database models.

Generate frontend pages.

Generate authentication system.

12. Debugging Using AI

After project generation:

Run the project.

Fix errors using AI.

Example prompt:

The backend API throws an error.

Here is the code.

Fix the bug and improve security.

AI will usually fix the issue quickly.

13. Final Workflow

Idea



Documentation



AI Code Generation



Testing



Debugging



Deployment

This workflow allows developers to build MVP products extremely fast.